

DataFrames with pandas, Polars and Arrow

pandas and Polars both give you a DataFrame. What is Arrow then?

YOUR PRESENTATION (15 MINUTES + 10 MIN Q&A)

This could be the structure of your talk. Nonetheless, you are free to improve or modify to ensure the topic will be more accessible.

1. The MeteoSuisse measurements are a table of typed columns, and a DataFrame is that table plus the operations to filter, join and aggregate it
2. pandas runs each step as you wrote it, so filtering to one station after joining the station metadata reads every station first
3. Polars can read the whole chain before running it and push that filter down into the read, so most of the file never loads
4. Arrow is not a third DataFrame, it is the memory layout under both, so the same MeteoSuisse table passes from one to the other without a copy
5. The demo runs one MeteoSuisse aggregation in each, times it honestly, then hands the table across through Arrow

YOUR DEMO (AIM AT 2 MIN, BUT FLEXIBLE IF YOU BLEND IT NICELY WITH THE PRESENTATION)

The same small transformation in pandas and in Polars. Show the eager versus lazy difference, with one honest measurement. Then show the two handing data to each other through Arrow without copying it.

Demo should run in about 2 minutes, and should be a demo unless a teacher allows you to bring pre-baked results.

YOUR THREE DELIVERABLES

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| C1 Angle — one slide: the topic in your own words, the question you answer, what your demo will show | Wednesday 23 September |
| C2 Demo runs — the code, a README, one command | Friday 25 September |
| C3 Deck — complete slides, demo embedded | Sunday 27 September |

All three interim materials must be uploaded into ISC Learn by 18:00; the demo as a repository link. These deliverables are mandatory, and the teacher will provide feedback to improve your work.

READING

McKinney (2010). Data structures for statistical computing in Python [doi:10.25080/Majora-92bf1922-00a](https://doi.org/10.25080/Majora-92bf1922-00a)
 Ritchie Vink. Polars: DataFrames in the multi-core era (PyData NYC 2023) www.youtube.com/watch?v=tqcudsykOGc
 A deep dive into the Arrow Columnar format with pyarrow and nanoarrow www.youtube.com/watch?v=k0mkDXWfLb4

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